

Applied Robotics

EE52043

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Lorenzo Giunta

Post-doctoral Researcher



- Additive manufacturing
 - D4AM, Material properties of AM, Responsive AM
- Human Centred Design
- Medical Robotics



Zuhayr Rymansaib

Post-doctoral Researcher



- Previous research: 3D printed conductors
- Current work:
 - Autonomous marine robotics,
 - Underwater acoustics,
 - Transducer design and signature acquisition



Contents

ILOs

What we expect
you to have
learned and be
able to show at
the end

Task

Your applied
robotics design
task

Groups

The groups you
will be working
with

Set-up

Getting Unity and
the simulator up
and running on
your PCs

Learning Outcomes

1. Apply a comprehensive knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex robotic systems.



Learning Outcomes

2. Formulate and analyse complex problems in robotic systems. This will involve evaluating available methods in robotics hardware, software and platforms and formulating solutions to address real-world problems, discussing the limitations of the techniques employed



Learning Outcomes

3. Select and apply appropriate computational and practical techniques to model complex problems, applied to the design and control of robotic engineering solutions, discussing the limitations of the techniques employed

Learning Outcomes

4. Apply an integrated or systems approach to the solution of complex robotic problems

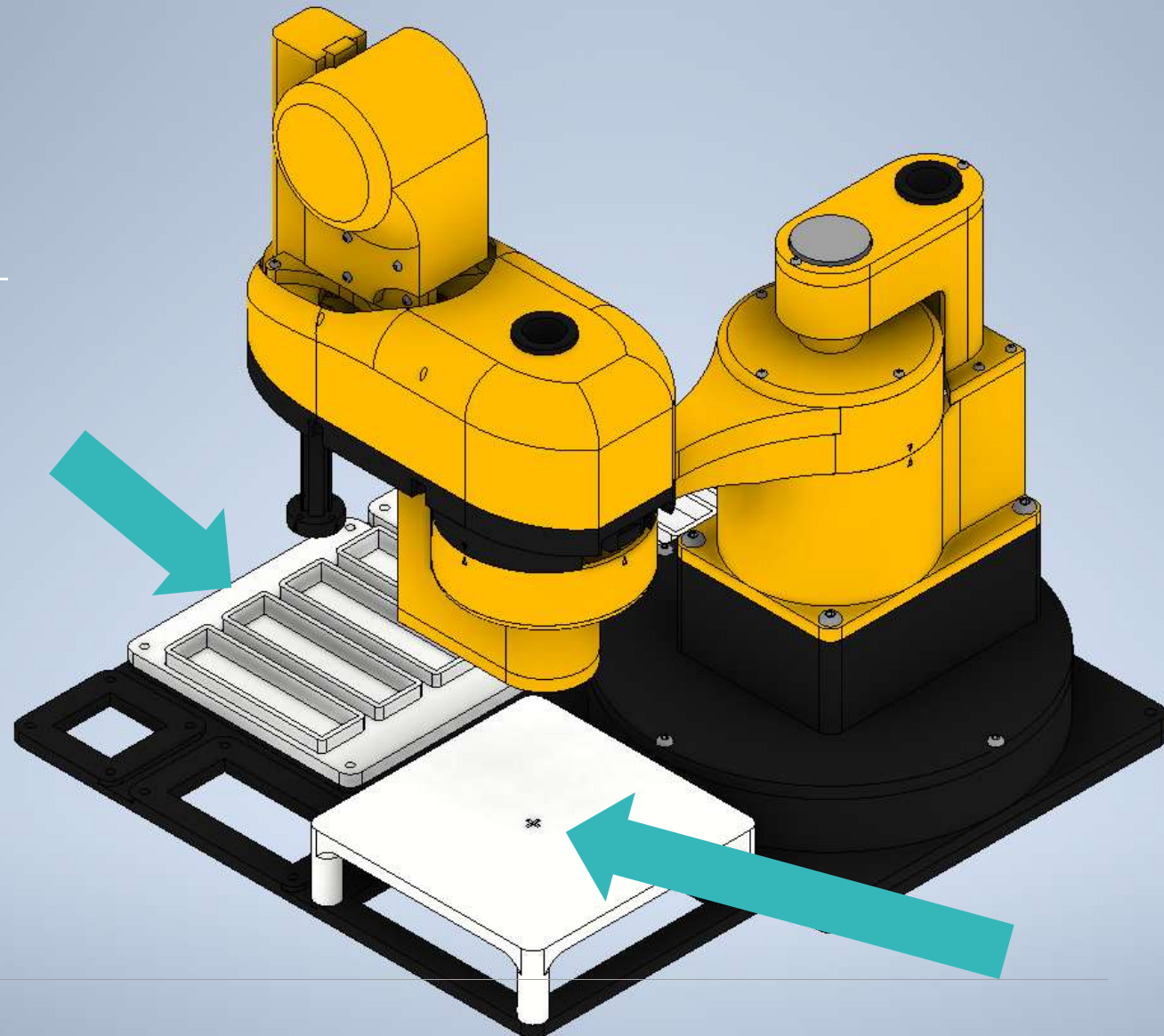


Design Task

- Combination of pick and place and sorting
- You will be expected to sort a random set of resistors into separate containers, grouping them correctly
- They will all be through hole resistors

Design Task

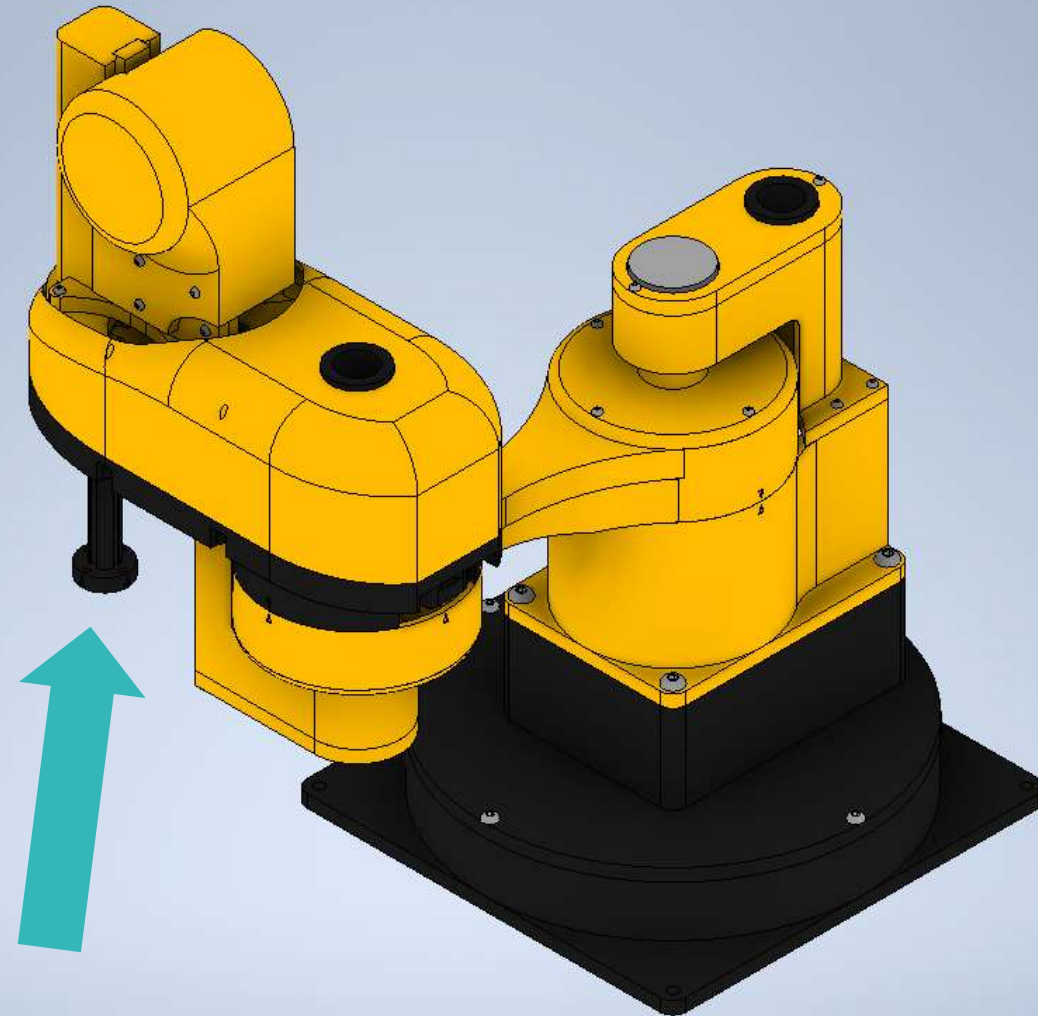
- Resistors will be placed on this table
- And will then need to be sorted into these trays



Design Task

- The robot, table, trays, and base will all be provided
- You will need to design the gripper however

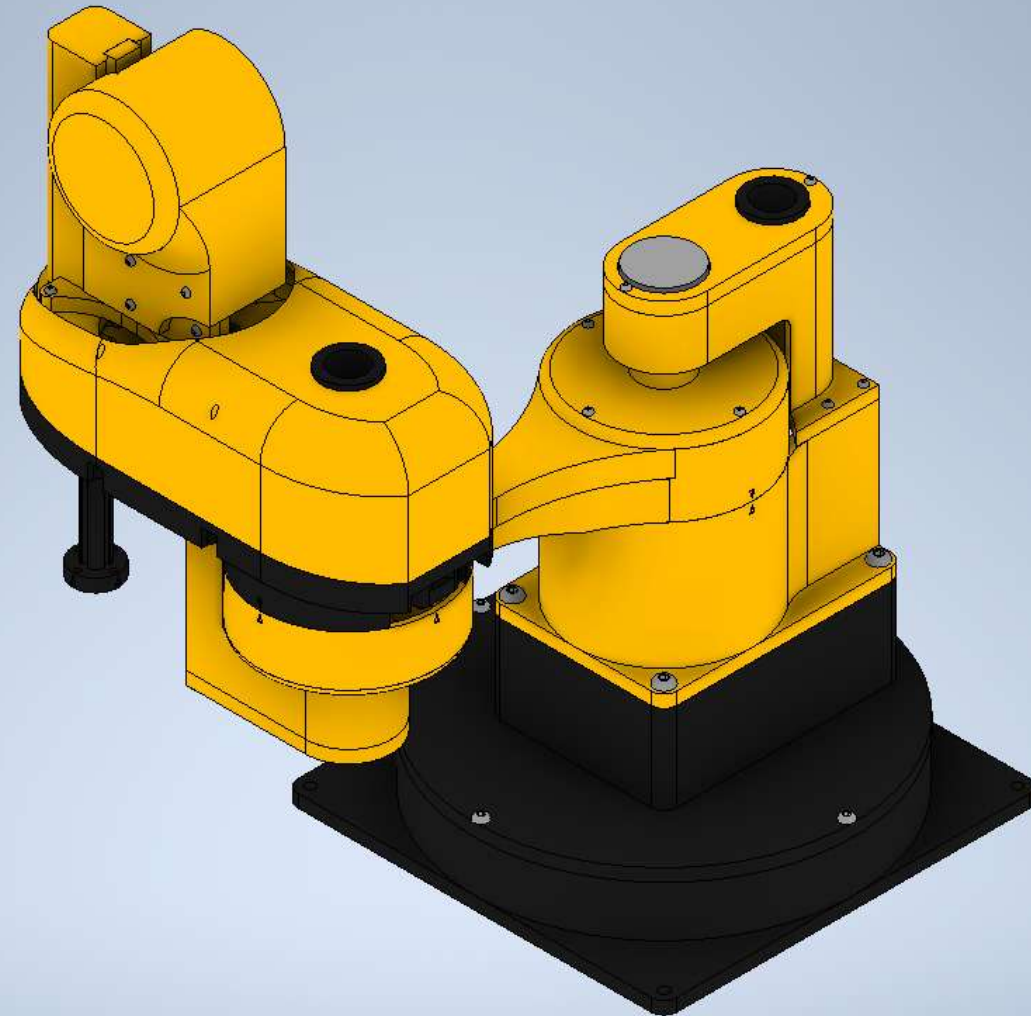
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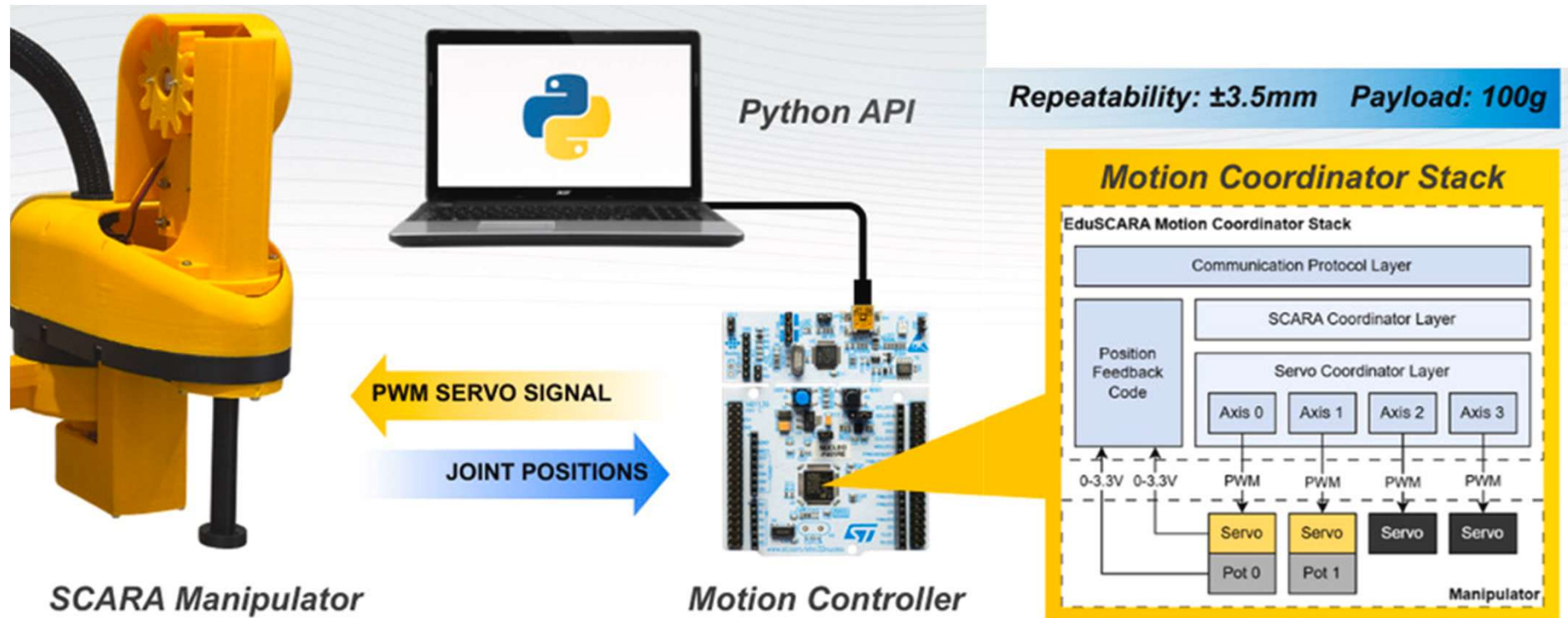


Design Task – SCARA Robot

- The robot is a Selective Compliance Assembly Robot Arm (SCARA)
- Commonly used for pick and place operations in industry

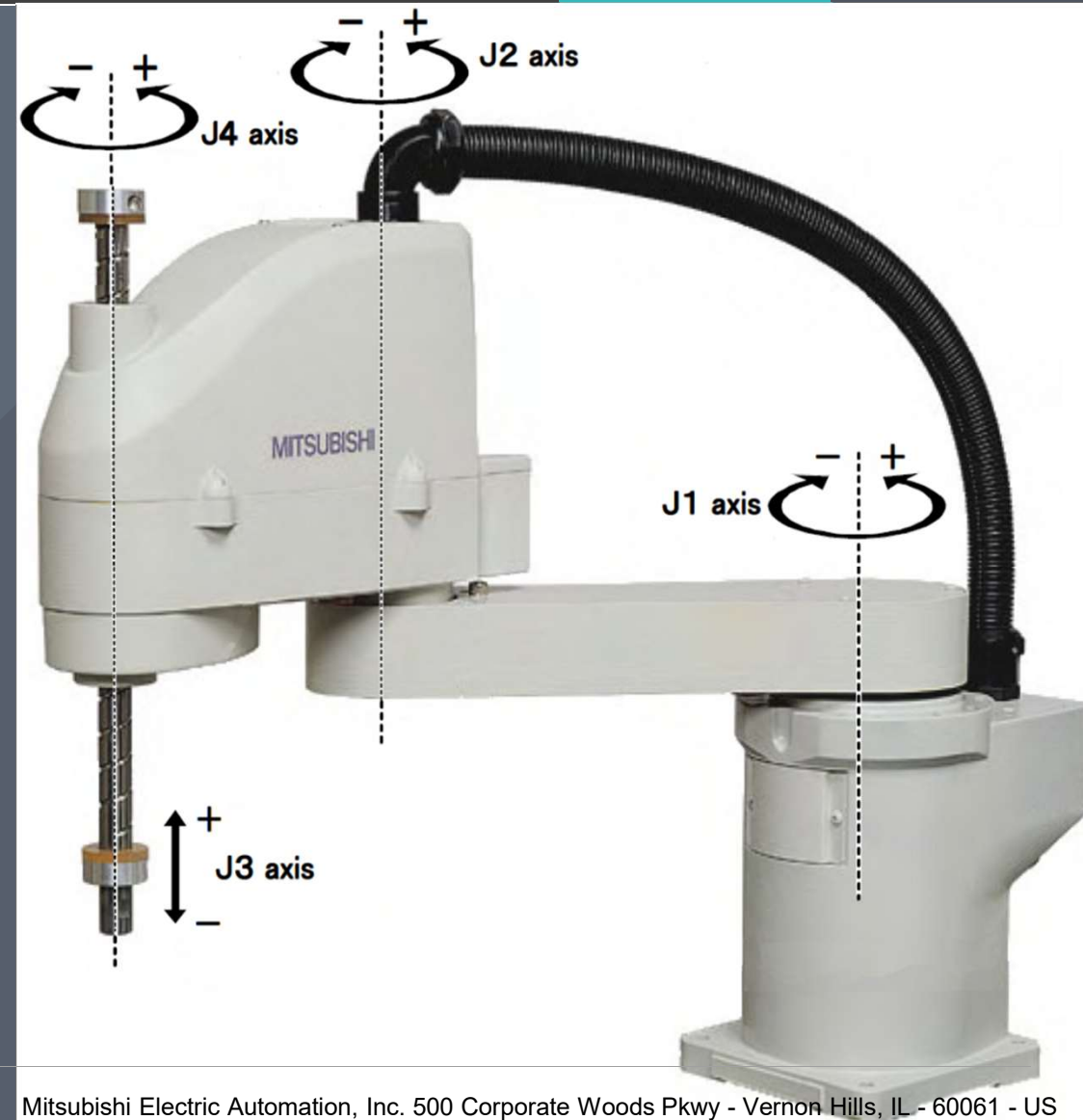
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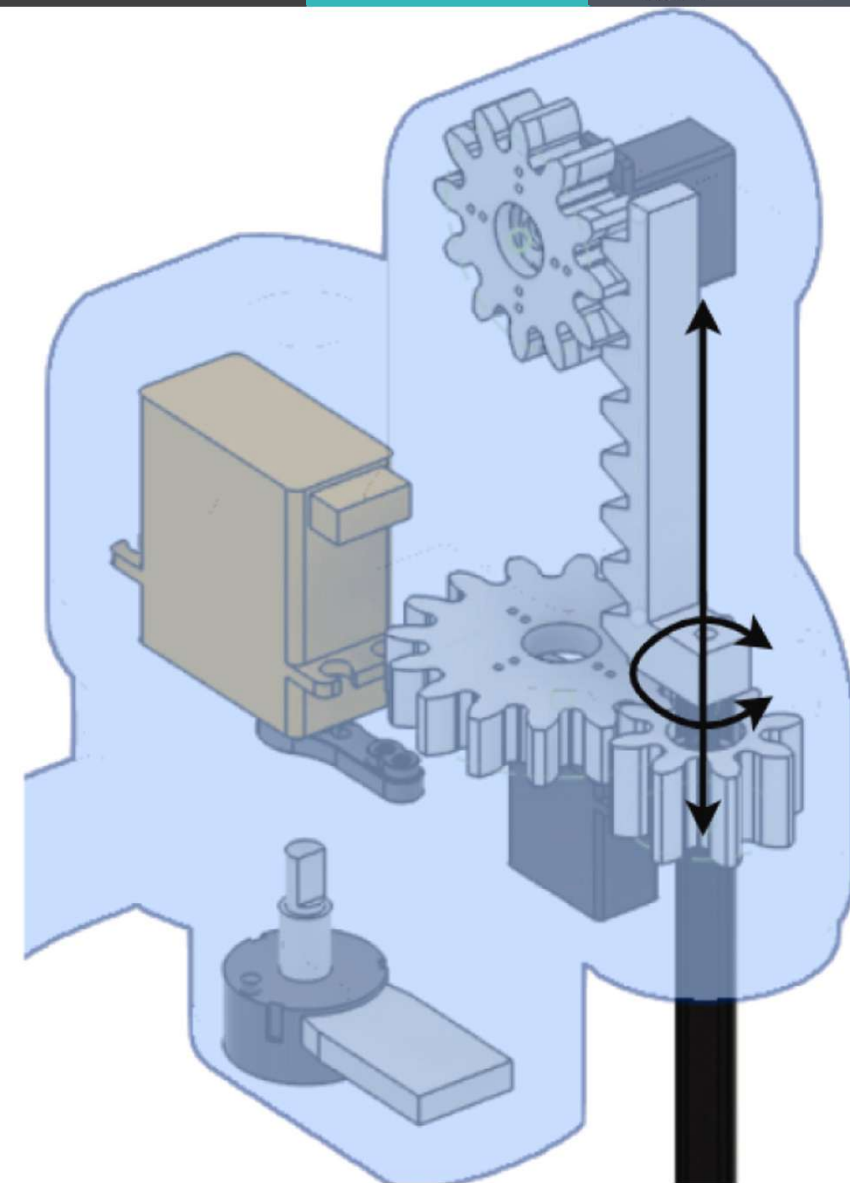
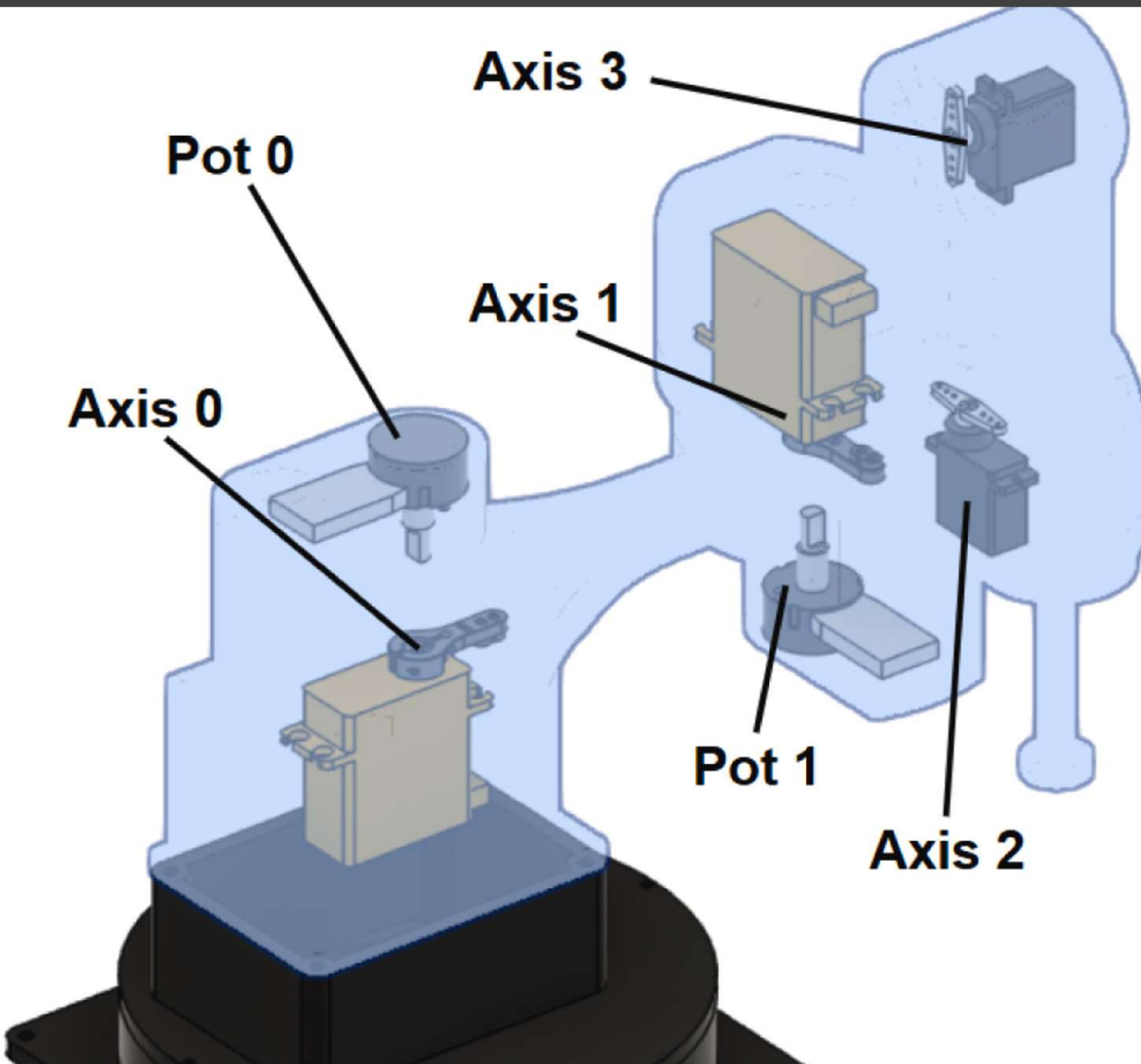


Design Task – SCARA Robot

- 4 degrees of freedom (excluding gripper)
- Rigid in the z axis
- Degrees of motion:
 - J1: 180°
 - J2: 180°
 - J3: 10 cm
 - J4: 360°



Mitsubishi Electric Automation, Inc. 500 Corporate Woods Pkwy - Vernon Hills, IL - 60061 - US



Grading Criteria

Gripper Design

- Well matched to the task
- Considers compliance & reliability
- Demonstrates considerations of iterations and testing

Resistor Identification Strategy

- Integrates sensing, perception and decision-making
- Considerations for uncertainty & environmental variation
- Justifies technical choices

Grading Criteria

Robot Control, Speed, and Performance

- Precision, reliability and efficiency
- Appropriate safety considerations taken
- Speed, accuracy, robustness and repeatability

Task Completion and Reliability

- Reliable completion during demonstration
- No human intervention
- Consistent reproducible results

Grading Criteria

Robustness and Error Recovery

- Can handle variation, and unexpected conditions
- Responds automatically
- Shows consideration of failure modes

Deliverables

- Pre-recorded presentation to be submitted before
- Live demo of the robot working with Q&A based on presentation
- One single mark for SAMIS that combines the two outputs.

>



Questions?



Schedule

WK	WK date	Monday	Tuesday
1	28/9/26	1W 2.104 – Introduction to the unit	WH 1 LT – Gripper Design
2	5/10/26	2E 2.14 – First Lab Session	WH 1 LT – Gripper Design
3	12/10/26	2E 2.14 – Second Lab Session	WH 1 LT – Vision and Image Processing
4	19/10/26	2E 2.14 – Third Lab Session	WH 1 LT – Vision and Machine Learning
5	26/10/26	1W 2.104 – Kinematics overview	WH 1 LT – Motion sequencing and Calibration
6	2/11/26	1W 2.104 – Guest Session/Q&A	WH 1 LT – Guest Session/Q&A
7	9/11/26	2E 2.14 – Fourth Lab Session	WH 1 LT – Closed Loop Control
8	16/11/26	2E 2.14 – Fifth Lab Session	WH 1 LT – Understanding Failure Modes
9	23/11/26	2E 2.14 – Sixth Lab Session	WH 1 LT – Error Correction and Tolerances
10	30/11/26	2E 2.14 – Seventh Lab Session	WH 1 LT – Q&A session
11	7/12/26	2E 2.14 – Demos	WH 1 LT – Demos

Schedule

- You are expected to do some work outside of **these sessions** (about twice the amount of the taught hours)



Your Experiences

- Only 10 robots so you will work in groups. Groups also distribute skillsets and course backgrounds as not all of you have covered the same modules so far.
- We have tried to split you equally into groups of different backgrounds
- While the assessment is individual, we expect you to collaborate and share knowledge



Tips and Suggestions

- Be careful with the robots!
 - If you break them you will need to print replacement parts (CAD files in Repo)
- Try to use GitHub or similar for version control
- We strongly recommend you use python for this project
- Share knowledge and help each other in your groups: play to your strengths
- Do not leave it to the last minute to let us know something has gone wrong

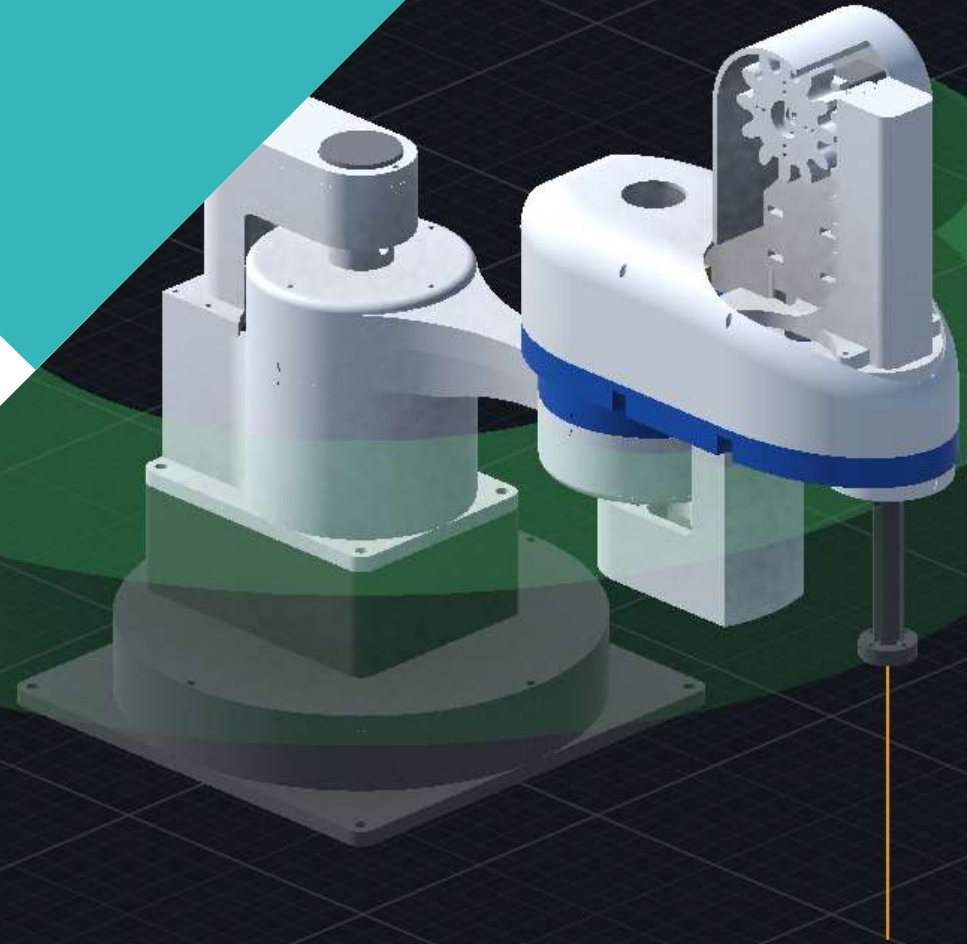


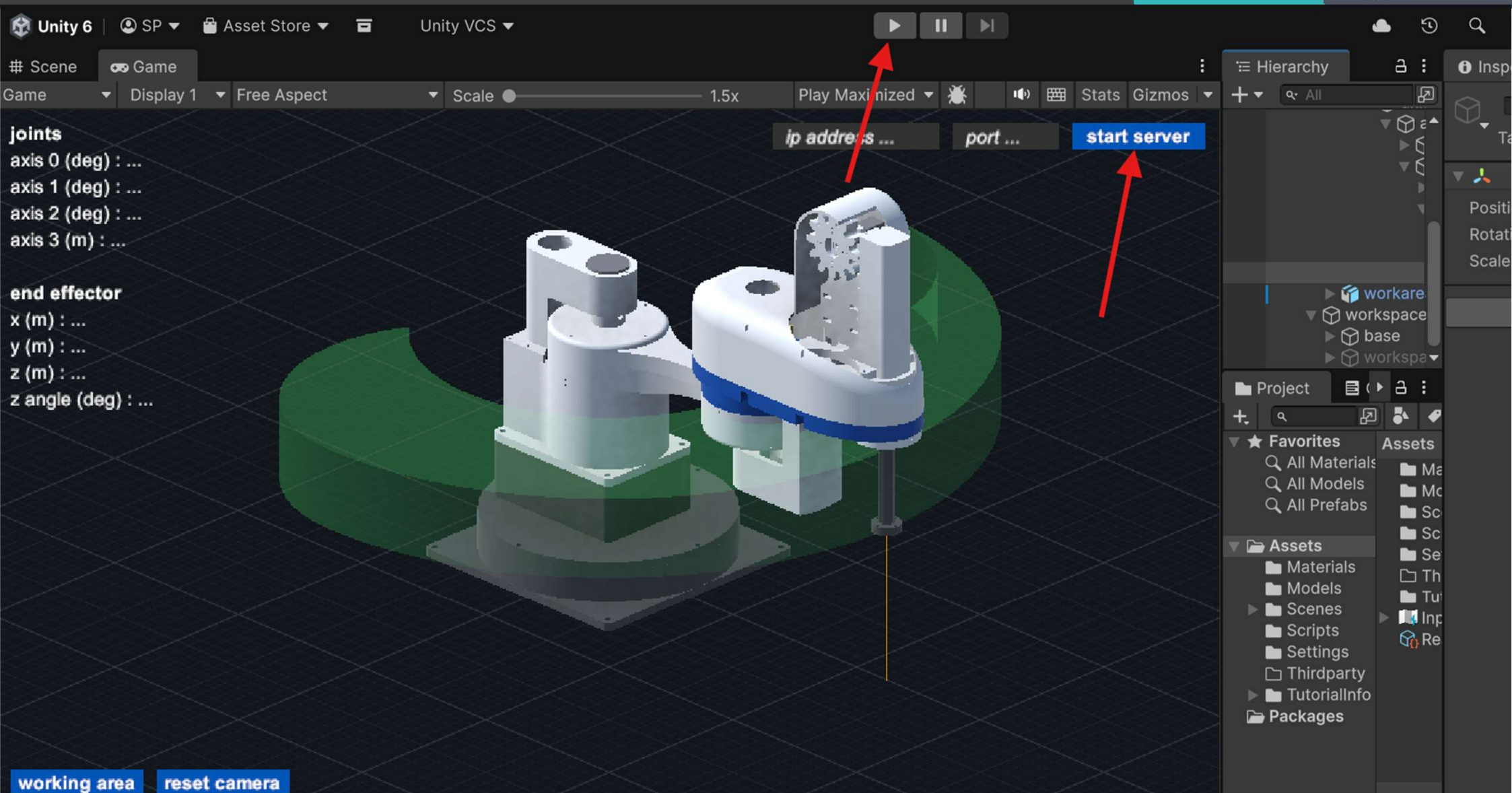
Questions?



Installing the Simulator

- Download Unity 6000.3.24f
- Go to the GitHub repo
- Download the `sim_scara_motion_controller_api` and put it in the project folder where you will keep your python files
- Download `mini_scara_simulator`





```
% Create an instance of the API class
scara = sim_scara_motion_controller;

% Test move joint command
scara.MoveJoint(0,90,1);
scara.MoveJoint(0,0,1);
scara.MoveJoint(1,90,1);
scara.MoveJoint(1,0,1);
scara.MoveJoint(2,90,1);
scara.MoveJoint(2,0,1);
scara.MoveJoint(3,0.075,1);
scara.MoveJoint(3,0.025,1);

% Test move joints command
scara.MoveJoints(90,30,90,0.05,1);

% Test move coord command
scara.MoveCoord(0.1,0.1,90,0.05,1);
```

Command	Inputs	Description	Outputs
Initialise	Link1: Horizontal distance between axis 0 and 1 in metres. Link2: Horizontal distance between axis 1 and 2 in metres. Z_min: Minimum z coordinate physically achievable in metres. Z_max: Maximum z coordinate physically achievable in metres. Settling Time: Time allocated after each movement for the Eduscara to correct for under/overshoot. P_0: Proportional gain of axis 0. I_0: Integral gain of axis 0. D_0: Differential gain of axis 0. P_1: Proportional gain of axis 1. I_1: Integral gain of axis 1. D_1: Differential gain of axis 1.	Initialise function defines physical dimensions and limitations of the robot and allows for PID tuning in Axis 0 and 1. Also defines the settling time between each movement command.	none
Move Joint	Axis: Axis to be rotated (0 to 3). Angle: Angle to rotate to (deg). If axis is 3 then Angle is a Z coordinate (m) Time: Time taken to complete the movement in seconds.	Rotates an axis to a specified angle in a specified time.	none
Move Joints	Angle0: Angle of Axis 0 (deg). Angle1: Angle of Axis 1 (deg). Angle2: Angle of Axis 2 (deg). Angle3: Z coordinate (m). Time: Time taken to complete the movement in seconds.	Rotates all Axes to specified angles in specified time.	none
Move Coord	X: X coordinate to move to (m). Y: Y coordinate to move to (m). Z_angle: Angle of end effector in the Z dimension (deg). Z: Z coordinate to move to (m). Time: Time taken to complete the movement in seconds.	Moves the end effector to a specified location and rotation in a specified time.	none